The University of Texas at Austin
In-Class Work 18
M339D Introduction to Financial Mathematics

November 09, 2026

Instructions: Provide your complete solution to the following problems. Final answers only, without appropriate justification, will receive zero points even if correct.

Log-Normal Stock Prices: Tail Probabilities.

Motivation.

Problem 18.1: *Let the stock prices be modeled using the lognormal distribution. Under the risk-neutral probability measure, the mean stock price at time—1 equals 120 and the median stock price 115. What is the risk-neutral probability that the time—1 stock price exceeds 100?*

→:

$$\mathbb{P}^*[S(1)>100] = ?$$

$$\mathbb{P}^*\left[S(0)\,e^{(r-\frac{\sigma^2}{2})\cdot 1+\sigma\sqrt{1}\cdot Z}>100\right]=$$

$$=\mathbb{P}^*\left[\underbrace{S(0)\,e^{(r-\frac{\sigma^2}{2})\cdot 1}}_{\text{median of } S(1)}\cdot e^{\sigma\cdot Z}>100\right]=$$

$$\text{median of } S(1) = 115$$

$$=\mathbb{P}^*\left[115\cdot e^{\sigma\cdot Z}>100\right]=$$

$$=\mathbb{P}^*\left[e^{\sigma Z}>\frac{100}{115}\right]=\mathbb{P}^*\left[Z>\frac{1}{\sigma}\ln\left(\frac{100}{115}\right)\right]$$

$$\frac{\text{mean}}{\text{median}}=\frac{S(0)e^{r\cdot 1}}{S(0)e^{(r-\frac{\sigma^2}{2})\cdot 1}}=e^{\frac{\sigma^2}{2}}\Rightarrow$$

$$\Rightarrow \sigma=\sqrt{2\cdot\ln\left(\frac{120}{115}\right)}=0.2918$$

$$\mathbb{P}^*[Z>-0.48]=N(0.48)=0.6844$$

□

Derivation.

Problem 18.2: *Consider a non-dividend-paying stock. What is the probability that the stock outperforms a risk-free investment* under the risk-neutral probability measure $\mathbb{P}^*$?

→: The initial investment amount is $S(0)$.

- If risk-free ⇒ @ time T the balance is $S(0)e^{rT}$
- If stock ⇒ the wealth is $S(T)$

$$\mathbb{P}^*[S(T) > S(0)e^{rT}] = ?$$

This question is the same as if we asked whether the stock investment has a positive profit.

In the Black-Scholes model:

$$S(T) = S(0)e^{(r-\frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z} \quad \text{w/ } Z \sim N(0,1)$$

$$\mathbb{P}^*[S(0)e^{(r-\frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z} > S(0)e^{rT}] =$$

ln( ) ln( ) ln(·) increasing

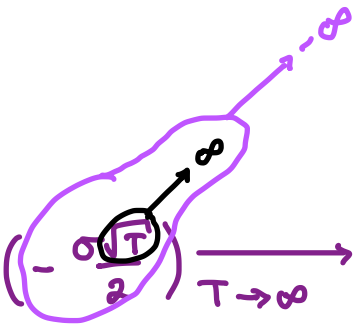
$$= \mathbb{P}^*[(r-\tfrac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z > rT]$$

$$= \mathbb{P}^*[-\tfrac{\sigma^2}{2}\cdot T + \sigma\sqrt{T}\cdot Z > 0]$$

$$= \mathbb{P}^*[\sigma\sqrt{T}\cdot Z > \tfrac{\sigma^2}{2}\cdot T]$$

$$= \mathbb{P}^*[Z > \tfrac{\sigma\sqrt{T}}{2}]$$

$$= \mathbb{P}^*[Z < -\tfrac{\sigma\sqrt{T}}{2}] = N\left(-\tfrac{\sigma\sqrt{T}}{2}\right) \xrightarrow[T\to\infty]{} 0$$



□

Problem 18.3: *Consider a non-dividend-paying stock and European call option on that stock. As usual, let K denote the option's strike price and let T denote its exercise date.* Under the risk-neutral probability measure $\mathbb{P}^*$, *what is the probability that the call is* ***in-the money*** *at time-T?*

$$\mathbb{P}^*[S(T) > K] =$$

$$= \mathbb{P}^*\left[S(0)\, e^{(r-\frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z} > K\right]$$

$$= \mathbb{P}^*\left[e^{(r-\frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z} > \frac{K}{S(0)}\right] = \quad \ln(\cdot) \text{ increasing}$$

$$= \mathbb{P}^*\left[(r-\tfrac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z > \ln\left(\frac{K}{S(0)}\right)\right]$$

$$= \mathbb{P}^*\left[\sigma\sqrt{T}\cdot Z > \ln\left(\frac{K}{S(0)}\right) - (r-\tfrac{\sigma^2}{2})\cdot T\right]$$

$$= \mathbb{P}^*\left[Z > \frac{1}{\sigma\sqrt{T}}\left[\ln\left(\frac{K}{S(0)}\right) - (r-\tfrac{\sigma^2}{2})\cdot T\right]\right]$$

Symmetry of $N(0,1)$

$$= \mathbb{P}^*\left[Z \leq \frac{1}{\sigma\sqrt{T}}\left[\ln\left(\frac{S(0)}{K}\right) + (r-\tfrac{\sigma^2}{2})\cdot T\right]\right]$$

$$=: d_2$$

$$\boxed{\mathbb{P}^*[S(T) > K] = N(d_2)} \quad \leftarrow$$

□

Consequently.

The probability under $\mathbb{P}^*$ that the otherwise identical put is in-the-money @ time-T is

$$\mathbb{P}^*[S(T) < K] = 1 - N(d_2) = N(-d_2)$$

Numerical Exercises.

Problem 18.4: *Assume the Black-Scholes model. The current stock price is* 50 *and the stock's volatility is* 0.25. *The stock pays no dividends. The continuously compounded risk-free interest rate is* 0.05. *Consider a European call option on this stock with strike price* 55 *and exercise date in six months.* Under the risk-neutral probability measure $\mathbb{P}^*$, *what is the probability that the call is* ***in-the-money*** *at expiration?*

→: $\mathbb{P}^*[S(T) > K] = N(d_2)$

w/ $d_2 = \frac{1}{\sigma\sqrt{T}}\left[\ln\left(\frac{S(0)}{K}\right) + (r - \frac{\sigma^2}{2})T\right] =$

$= \frac{1}{0.25\sqrt{1/2}}\left[\ln\left(\frac{50}{55}\right) + (0.05 - \frac{0.25^2}{2})\cdot\frac{1}{2}\right] =$

$= \dots = -0.49$

$N(-0.49) = 0.3121$ □

Problem 18.5: *Assume the Black-Scholes model. The current stock price is 80 and the stock's volatility is 0.30. The stock pays no dividends. The continuously compounded risk-free interest rate is 0.05. Consider a one-year, 75-strike European put option on this stock.* Under the risk-neutral probability measure $\mathbb{P}^*$, *what is the probability that the put is* ***in-the-money*** *at expiration?*

$\longrightarrow$:

$$\mathbb{P}^*[S(T) < K] = N(-d_2)$$

w/ $$d_2 = \frac{1}{0.3\sqrt{1}}\left[\ln\left(\frac{80}{75}\right) + \left(0.05 - \frac{0.3^2}{2}\right)\cdot 1\right]$$

$$= 0.23$$

$$N(-0.23) = 0.4090$$

□